

Fashion Design with FlexTruss Approach

Lingyun Sun
Zhejiang University, China

Jiaji Li
Zhejiang University, China

Danli Luo
Carnegie Mellon University, USA

Ziqi Fang
Zhejiang University, China

Yitao Fan
Zhejiang University, China

Zhi Yu
Zhejiang University, China

Yu Chen
Zhejiang University, China

Deying Pan
Zhejiang University, China

Yue Yang
Zhejiang University, China

Jianzhe Gu
Carnegie Mellon University, USA

Yijun Zhao
Zhejiang University, China

Lining Yao
Carnegie Mellon University, USA

Ye Tao
Zhejiang University City College,
China

Guanyun Wang*
Zhejiang University, China

ABSTRACT

This demo presents FlexTruss, a design and construction pipeline based on the assembly of modularized truss-like objects fabricated with conventional 3D printers and assembled by threading. To create an end-to-end system, a parametric design tool with an optimal Euler path calculation method is developed, which can support both inverse and forward design workflow and multi-material construction of modular parts. In addition, the workflow guides the assembly of printed truss modules by threading. Finally, a series of application cases to demonstrate the affordance of FlexTruss are presented. We believe that FlexTruss extends the design space of 3D printing beyond typically hard and fixed forms, and it will provide new capabilities for designers and researchers to explore the use of such flexible truss structures in human-object interaction.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interactive systems and tools.

KEYWORDS

Mesh structures, Shape-changing interfaces, personal fabrication, 3D printing, Euler path

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*Corresponding authors: Guanyun Wang, guanyun@zju.edu.cn

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1 INTRODUCTION

In the Human-Computer Interaction (HCI) community, truss structures are widely used for prototyping since it is stable, lightweight, and saves material [1–3, 8]. Similarly, more researchers attempt to integrate truss structures with 3D printing to accelerate design iterations while maintaining stability [5, 11]. Additionally, printing objects with 2D truss-shaped patterns then morphing to 3D structures has also been demonstrated in 4D printing related works [9, 10]. However, 3D printed models are difficult to modify physically as they are supposed to be a truthful resemblance of the input shape and are not designed for continuous, interactive fabrication (Figure 1). As a result, a growing interest in developing real-time design, fabrication, and modification methods that allow users to receive continuous feedback is revealed by recent works [6, 7, 12].

In this demo, we present FlexTruss, a design and fabrication pipeline that use a truss-like structure to approximate 3D designs across scales. Our method allows on-demand modifiable prototyping with modularized 3D printed components assembled by simple threading. Each individual node on the FlexTruss structure is hollowed to incorporate a threading wire that traverses all nodes. A design tool, powered by optimal Eulerian circuit calculation coupled with multi-node generation and sorting, is developed to aid this process. We also experiment with various structures and multi-material systems to expand the physical properties and malleability of the assembled models for different uses. While our approach requires some manual assembly after the computer-aided design and 3D printing processes, it facilitates intuitive, instructive, and sensible design by allowing users to reflect on and tinker with the final artifact in their hands. We believe that FlexTruss extends the design space of 3D printing beyond typically hard and fixed forms, and it will provide new capabilities for designers and researchers to explore the use of such flexible truss structures in human-object interaction.

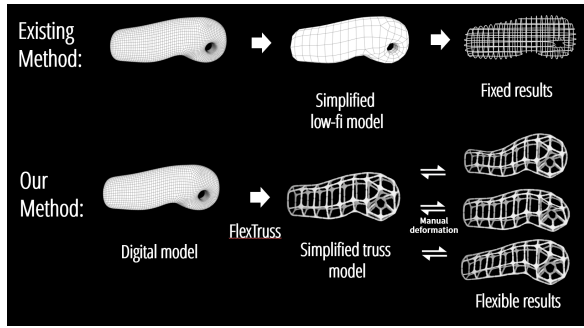


Figure 1: FlexTruss method for modifiable prototyping.

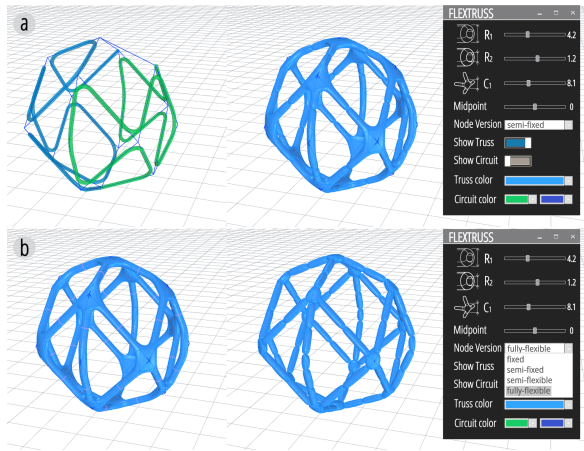


Figure 2: FlexTruss interface. (a) Users can visualize the threading or (b) change node types for different flexibility requirements.

2 COMPUTATIONAL DESIGN

2.1 User Interface

Using Grasshopper, a parametric design plugin for Rhinoceros, we developed a user interface (Figure 2) for FlexTruss to allow users to customize node types, as they have different degrees of freedom, and visualize the threading path. It also allows users to input models to create case-specific solutions. The interface also allows users to change the size of the truss (inner diameter R1 to accommodate threading wire's size, outer diameter R2 to determine the pipe size, and node size C1).

2.2 Demonstrating the Inverse Design Process

We demonstrate how to use our design tool to inversely design and fabricate a Stanford Bunny. Firstly, after the user imports a 3D model (Figure 3a), the system simplifies the model into a truss mesh structure, with the density of the mesh being one of the adjustable parameters (Figure 3b). After previewing the simplified result, the user could tune the size of nodes on demand and check the node generation (Figure 3c). Meanwhile, the system generates a single path of connected pipes following a calculated Euler path that traverses all nodes (Figure 3d). The radius of the pipe varies

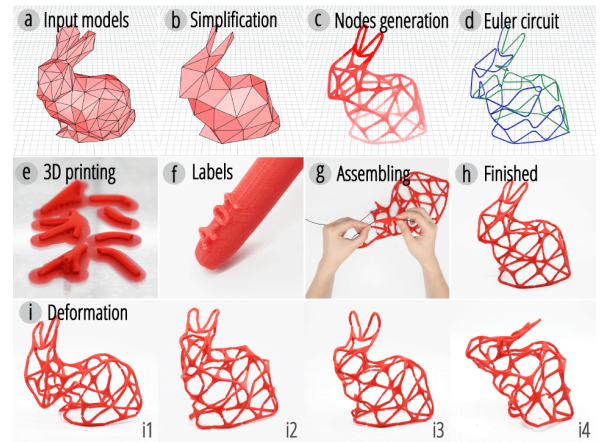


Figure 3: Inverse design workflow

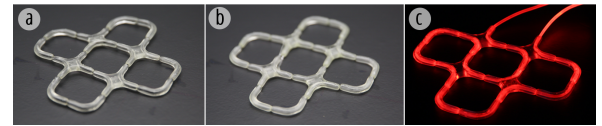


Figure 4: Threading results of (a) a PVC coated galvanized wire; (b) an elastic rope; (c) an optical fiber.

based on the material of the wire. Eventually, the multi-directional joints are generated and labeled (Figure 3e, f). After making the joints (e.g., by filament deposition modeling), users can assemble the FlexTruss prototype following the labels (Figure 3g). Lastly, the user can manipulate the prototype with countless possibilities of deformation (Figure 3h).

3 MULTI-MATERIAL FABRICATION

For the 3D printing setup, we employ an Ultimaker S5 desktop 3D printer with default settings - 0.15 mm layer thickness and 30% infill with triangular pattern - to print the nodes with a common thermoplastic material.

Compared with the work Eulerian Wires [4], the materials used for our prototype are not limited to metal wire, which makes this method suitable for a wide range of easy-to-cut materials, such as wires, threads, and fibers, with suppleness, strength, and even elasticity. The suppleness allows bending of FlexTruss objects without cracking. It also requires certain material strength that can support 3D shapes. Besides, certain elasticity would the objects to be stretched and worn as clothing.

More wiring and threading materials can be integrated with FlexTruss. Figure 4 shows three selected materials with different properties. Among these materials, the PVC coated galvanized wire is used in a wide variety of applications including packaging, shaping, and fencing. It is also easily deformable, maintains the shape after manual manipulation, and cheap (less than \$0.02 per meter).

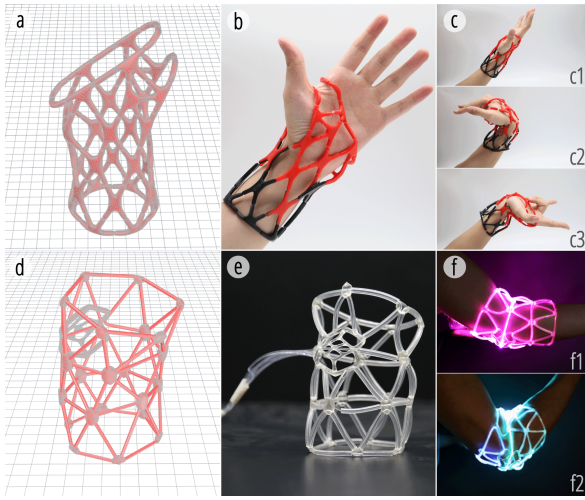


Figure 5: (a, d) Virtually generated nodes for wearable devices: the FlexTruss wristband (b, c) can adapt to the user's hand gesture, and the FlexTruss armband (e, f) can give color-changing feedback using an optical fiber as the threading wire.

4 MULTI-USE FASHION DESIGN EXAMPLES

To situate FlexTruss in a range of use cases in fashion design, we implement four wearable examples.

4.1 Wrist/Arm Guard Design

In this example, we thread FlexTruss with elastic rope to demonstrate a wearable device design. In Figure 5a-c, we complete the prototype by reconstructing a wristband with 40 nodes. The artifact can be worn by users with different wrist sizes and deformed according to the user's action. We also demonstrate that by replacing the nodes with more flexible nodes the same design can be transformed into an armband (Figure 5d-g). By embedding an optical fiber, this prototype can give real-time feedback on the wearer's gesture.

4.2 Jewelry Design

The accuracy of conventional FDM printers limits the size of nodes. Thus, we adopt stereolithography (SLA) printing to explore the minimum size of FlexTruss artifacts. In Figure 6a, we complete a flexible ring by reconstructing it with ten 10x6x2mm nodes. Figure 6b shows that the FlexTruss ring can be stretched and worn on fingers of different sizes, and the FlexTruss bracelet is stretchable PVC designed for superior comfort with a monolithic construction. Additionally, the shape, color, and flexibility can be easily augmented in the design process.

4.3 Handbag Design

The FlexTruss workflow empowers designers to create on-demand fashion pieces. Figure 7 shows an example of a multi-use handbag. Users can shape the handbag as functionally needed and even add straps to extend its functionality.

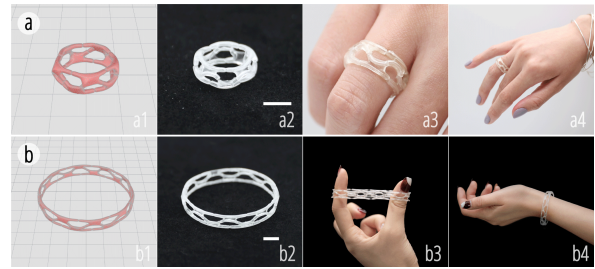


Figure 6: Jewelry prototype made by FlexTruss: (a) a flexible ring suitable for fingers of different sizes; (b) a FlexTruss bracelet could be stretched and worn without breakage. Scale bar: 10 mm

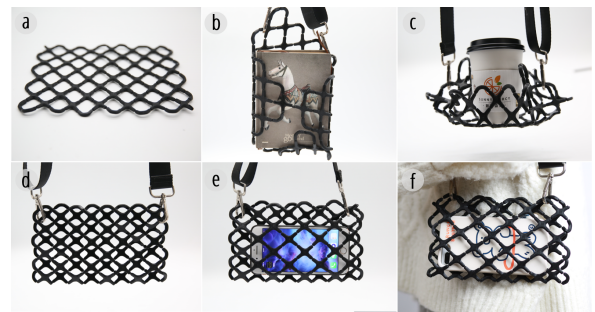


Figure 7: (a) A flat FlexTruss assembly can be transformed into (b) an envelope-shaped bag or (c) a cup holder. (d-f) adding straps to the same FlexTruss sheet will turn it into a shoulder bag.



Figure 8: Dress design enabled by FlexTruss: (a) the full assembly of the dress is made in three parts. The crop top (red) can be extended into a hip-length vest (pink), and a dress hem (crimson) can be added as well.

4.4 Dress Design

The boundless combinations of printing materials and threading wires enable multi-use design practice. This example showcases a piece of clothing (Figure 8a) consisting of three segments. A crop top (Figure 8b, red) can be extended into a hip-length vest (pink), both threaded by elastic ropes for maximum flexibility to suit the wearer's chest size. Moreover, the vest can be further extended with a metal wired FlexTruss hem (crimson). The metal wire acts like a pannier for an elegant flouncy look.

5 CONCLUSION

In this work, we have shown that threading compliant and flexible components brings the possibility of integrating multiple functions into one prototype for makers and designers using consumer-grade 3D printers. We developed a computational design tool to create truss structures, which can be directly 3D printed as well as easily assembled. The application examples made by our design tool showcased multi-material, multi-form, and multi-use applications with shape-changing, physical property-changing, and media embedding features. We look forward to extending our method into future sustainable fabrication.

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